

Identifying errors in proofs - Set 2 - Solutions

Specification Err1, Err2.

Answer key

Question	Answer	Topic
1	A	Identifying errors in proofs
2	A	Identifying errors in proofs
3	A	Identifying errors in proofs
4	A	Identifying errors in proofs
5	A	Identifying errors in proofs
6	E	Identifying errors in proofs
7	A	Identifying errors in proofs

Worked solutions

Question 1 ERR-07 · Identifying errors in proofs · spec Err2

A student writes the following argument.

I $0.1 > 0.01$.

II Taking logarithms to base 10 of both sides: $-1 > -2$.

III Multiplying both sides by $\log_{10} \frac{1}{2}$, which is negative, reverses the inequality:
 $-\log_{10} \frac{1}{2} < -2\log_{10} \frac{1}{2}$.

Which of the following is correct?

Answer: A

Logarithms are increasing functions, so line II is valid; $\log_{10} \frac{1}{2} < 0$, so reversing the inequality in line III is correct.

Question 2 ERR-08 · Identifying errors in proofs · spec Err1

Claim: if n is an integer and n^2 is divisible by 4, then n is divisible by 4.

I Suppose n^2 is divisible by 4.

II Then $n^2 = 4k$ for some integer k .

III So $n = 2\sqrt{k}$, and since n is an integer $\sqrt{k}$ is an integer.

IV Hence n is even, so $n = 2m$, and $n^2 = 4m^2$ is divisible by 4, so n is divisible by 4.

Which of the following is true?

Answer: A

$n = 2$: $n^2 = 4$ is divisible by 4 but n is not. Lines I-III are correct deductions (n even). Line IV concludes ' n divisible by 4' from ' n^2 divisible by 4', which is exactly the unproved claim - circular and false.

Question 3 ERR-11 · Identifying errors in proofs · spec Err1

Claim: $\int_{-1}^1 \frac{1}{x^2} dx = -2$.

I An antiderivative of x^{-2} is $-x^{-1}$.

II So the integral equals $[-\frac{1}{x}]_{-1}^1 = -1 - 1 = -2$.

III Hence the value is -2 .

Which of the following is correct?

Answer: A

The integrand is positive everywhere it is defined, so a negative answer is a warning sign; the function is not defined on the whole interval and the integral does not exist in the ordinary sense.

Question 4 ERR-13 · Identifying errors in proofs · spec Err1

Claim: for all positive real x , $x + \frac{1}{x} \geq 2$.

- I Suppose $x + \frac{1}{x} \geq 2$.
- II Multiply by $x > 0$: $x^2 + 1 \geq 2x$.
- III So $x^2 - 2x + 1 \geq 0$, i.e. $(x - 1)^2 \geq 0$, which is true.
- IV Hence the claim is proved.

Which of the following best describes the argument?

Answer: A

Starting from the conclusion and reaching a truth does not prove the conclusion unless every step is reversible and the argument is presented in the forward direction. Here the steps are reversible, so the repair is to write it backwards.

Question 5 ERR-15 · Identifying errors in proofs · spec Err1

Claim: the sum of the first n odd numbers is n^2 .

- I The first odd number is $1 = 1^2$, so the claim holds for $n = 1$.
- II Suppose the claim holds for some n , i.e. $1 + 3 + \dots + (2n - 1) = n^2$.
- III Adding the next odd number, $2n + 1$, gives $n^2 + 2n + 1 = (n + 1)^2$.
- IV So the claim holds for $n + 1$, and hence for all positive integers n .

Which of the following is correct?

Answer: A

This is a valid proof by induction: a base case, and a step showing that truth for n implies truth for $n + 1$.

Question 6 ERR-17 · Identifying errors in proofs · spec Err1

Claim: if $f'(a) = 0$ and $f''(a) = 0$ then f has a point of inflexion at $x = a$.

Which of the following shows the claim is false?

Answer: E

x^4 has $f'(0) = f''(0) = 0$ but a minimum at 0, not an inflexion. (x^3 does have an inflexion; x^2 has $f''(0) \neq 0$.)

Question 7 ERR-06 · Identifying errors in proofs · spec Err1

Claim: the function $f(x) = x^3 - 3x^2 + 3x$ is increasing for all real x .

- I $f'(x) = 3x^2 - 6x + 3 = 3(x - 1)^2$.
- II So $f'(x) \geq 0$ for all x , with equality only at $x = 1$.
- III Since $f'(x) > 0$ for all x , f is increasing.

Which of the following is true?

Answer: A

f is (strictly) increasing because $f'(x) \geq 0$ with equality only at an isolated point, so $f(a) < f(b)$ whenever $a < b$; but line III's assertion ' $f'(x) > 0$ for all x ' is false at $x = 1$. There is no turning point at $x = 1$ (the derivative does not change sign).